

Mathematics, 2015 EC, natural science, entrance exam, round 1

Grade 12 · Mathematics · 2015 E.C.

Time allowed: 120 minutes

Questions: 52

Total marks: 52

Name _____ Grade _____ ID No. _____ Date _____

1. A 20m high building casts a shadow of 4m, at the same time a 35m tree casts a shadow of x m. What is the value of x? (1)
 - A. 8m
 - B. 11m
 - C. 7m
 - D. 15m
2. A car started moving 9km to the East then moved 4 km to the North and then 12km towards West to reach its destination. How far the car it travelled from its starting point? (1)
 - A. 25km
 - B. 12km
 - C. 4km
 - D. 5km
3. When the polynomial $P(x) = 3x^6 + 5x^4 - 7x^3 + 2kx^2 + 3$ is divided by $(x + 1)$, the remainder is 4. What is the value of k? (1)
 - A. -5
 - B. -7
 - C. 3
 - D. 6
4. Which of the following statement is true about $f(x) = x^2 - 6x + 10$? (1)
 - A. $f(x) \geq 2$ for all x in the domain of f.
 - B. The line $x = 1$ is the axis of symmetry of the graph of f.
 - C. The graph of f opens downward.
 - D. The vertex of the graph of f is (3, 1).
5. What is the area of a regular hexagon inscribed in a circle of radius 6cm? (1)
 - A. $48\sqrt{3} \text{ cm}^2$
 - B. $54\sqrt{3} \text{ cm}^2$
 - C. 48 cm^2
 - D. $36\sqrt{3} \text{ cm}^2$
6. Which of the following defines a prime number? (1)
 - A. It is a number that has more than two positive factors.
 - B. It is a number whose only factor is 1.
 - C. It is a number that has only two positive factors.
 - D. It is a number whose only factor is itself.
7. Which of the following rational numbers is written in its standard form? (1)
 - A. 10×10^{-2}
 - B. 0.8×10^2
 - C. 0.09×10^5
 - D. 5×0^5

8. The solution set of the equation $3|2x - 4| = 6 - 2|1 - 5x|$ is (1)
- A. $\{\frac{1}{2}, \frac{5}{4}\}$ B. $\{-1, \frac{1}{2}, \frac{5}{4}\}$
 C. $\{\}$ D. $\{-1, \frac{1}{2}\}$
-
9. The domain of the function $f(x) = 2x^{\frac{2}{3}}$ is (1)
- A. $\mathbb{R} \setminus \{0\}$ B. $[0, \infty)$
 C. $\mathbb{R}$ D. $(0, 2)$
-
10. The slope of the line that makes an angle of 135° with the positive x-axis is equal to (1)
- A. 2 B. -1
 C. 1 D. 3
-
11. If $\tan 31^\circ = 0.81$ then what is the value of $\tan 509^\circ$ (1)
- A. 0.19 B. -0.81
 C. 0.81 D. -0.91
-
12. If $\sin \theta = -\frac{3}{5}$ and θ is a 4th quadrant angle, then what is the value of $\sec \theta$? (1)
- A. -5/3 B. 5/4
 C. 5/3 D. -5/4
-
13. If ABCD is a parallelogram with P a mid-point of $\overline{AB}$ and Q a mid-point of $\overline{CD}$, then which of the following is NOT necessarily true? (1)
- A. $\triangle ADQ \equiv \triangle CBP$ B. $\overline{AQ} \equiv \overline{PC}$
 C. APCD is a trapezium D. APCQ is a rhombus
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14. A regular pyramid has a hexagonal base whose side length is 12cm. What is the lateral surface area of the pyramid in square centimeters if the length of each lateral edge is 10cm? (1)
- A. 360 B. 72
 C. 288 D. 720
-
15. If $\begin{pmatrix} 2 & -1 \\ a & 4 \end{pmatrix} \begin{pmatrix} 1 & b \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -1 & 18 \end{pmatrix}$ Then what are the values of a and b respectively? (1)
- A. 3, 2 B. 2, -2
 C. -1, 2 D. 0, -1
-
16. Let $A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ -3 & 6 \end{pmatrix}$ Then A^{-1} is (1)
- A. $\begin{pmatrix} 1 & 0 & -3 \\ -1 & 2 & 6 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 0 \\ -1 & 2 \\ -3 & 6 \end{pmatrix}$
 C. $\begin{pmatrix} -1 & 2 & 6 \\ 1 & 0 & 6 \end{pmatrix}$ D. $\begin{pmatrix} -1 & 1 \\ 2 & 0 \\ 6 & -3 \end{pmatrix}$
-
17. On the set of Natural numbers $\mathbb{N}$, which one of the following propositions has a truth value True? (1)
- A. $(\exists x)(\exists y)(x - y = y - x)$ B. $(\forall x)(\forall y)(x \neq y \implies x > y)$
 C. $(\forall x)(\exists y)(x - y = 5)$ D. $(\forall x)(\exists y)(xy < x)$

18. In a certain month of the year, the temperature of a city was recorded and presented as follows: Temperature (in °C) 10-14 15-19 20-24 25-29 30-34 Frequency (number of days) 2 9 11 5 3 Which one of the following is the cumulative frequency of the fourth class? (1)
- A. 22 B. 27
C. 18 D. 20
19. A fair die is going to be rolled once, and let: E1 be the event that an even number occurs; E2 be the event that a prime number occurs; E3 be the event that a multiple of 3 occurs; E4 be the event that a multiple of 5 occurs; Then which of the following pairs of events are mutually exclusive? (1)
- A. E1 and E3 B. E3 and E4
C. E1 and E2 D. E2 and E4
20. The following is the distribution of the weight of 40 students in a certain class: Weight (in Kg) 40-48 49-57 58-66 67-75 76-84 f 8 18 7 5 2 Then for data given above, what is the median weight in kg? (1)
- A. 54.5 B. 55
C. 53 D. 53.8
21. A group of students, 5 from grade 10, 8 from grade 11 and 2 from grade 12 is to be formed. If a student is chosen randomly from this group, what is the probability that the chosen student is either from grade 10 or grade 12? (1)
- A. 7/15 B. 5/15
C. 12/15 D. 1/7
22. If the truth value of p is T, then which of the following compound propositions has a truth value T for any proposition q? (1)
- A. $\neg p \Leftrightarrow (p \vee \neg q)$ B. $(\neg p \wedge q) \Rightarrow q$
C. $(p \Rightarrow \neg p) \wedge q$ D. $(p \Rightarrow q) \wedge \neg p$
23. Which of the following functions is one to one? (1)
- A. $f = \{(1, 5), (2, 3), (5, 4), (6, 5)\}$ B. $f : (0, \infty) \rightarrow \mathbb{R}, f(x) = |x| - 1$
C. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 1$ D. $f = \{(x, y) \mid y \text{ is the mother of } x\}$
24. If $\frac{x^2+14}{(x+2)(x-1)^2} = \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$ then what are the values of A, B and C respectively? (1)
- A. 1, -1, 3 B. 3, -2, 6
C. 2, -2, -5 D. 2, -1, 5
25. Which of the following line is true about the intercept and symmetry of the graph of $f(x) = \frac{x^3-3x}{x^2-6}$? (1)
- A. The graph is symmetrical with respect to the y-axis.
B. Its y-intercept is 3.
C. Its x-intercepts are $\pm\sqrt{6}$.
D. The graph is symmetrical with respect to the origin.
26. Consider the lines $l1 : y = 2x + 3$ and $l2 : y = 1 - 3x$. Then the angle between $l1$ and $l2$ is (1)
- A. $\pi/4$ B. $3\pi/2$
C. $\pi/2$ D. $\pi/6$
27. The equation of the parabola with vertex V(2, -1) and focus F(-1, -1) is (1)
- A. $(x + 1)^2 = -12(y + 1)$ B. $(y + 1)^2 = -12(x - 2)$
C. $(y + 1)^2 = 12(x - 2)$ D. $(x - 2)^2 = -12(y + 1)$

28. What is the value of $\lim_{x \rightarrow 1} \frac{x^2-1}{x^2+1}$? (1)
- A. 3
B. 1
C. 2
D. 0
-
29. Which one of the following is the value of a so that the function given by $f(x) = \begin{cases} ax + 1 & \text{for } x \geq 1 \\ x^2 - 4 & \text{for } x < 1 \end{cases}$ is continuous at 1? (1)
- A. 4
B. -4
C. -1
D. 1
-
30. Let $f(x) = -\frac{1}{3} \cos\left(2x^3 - \frac{\pi}{3}\right) + 1$. Then what is the period and Phase shift of the graph of f respectively? (1)
- A. $4\pi/9$ and $2\pi/9$
B. $2\pi/3$ and $2\pi/9$
C. $3\pi/2$ and $4\pi/9$
D. $\pi/9$ and $4\pi/9$
-
31. What is the fifth term of the sequence $\{2^n + 4n - 5\}_{n=0}^\infty$? (1)
- A. 27
B. 17
C. 47
D. 15
-
32. Which one of the following numbers is the upper bounds of the sequence $\{\frac{2n+1}{n+1}\}_{n=1}^\infty$? (1)
- A. $3/2$
B. 0
C. 1
D. 4
-
33. Let the sequence $\{a_n\}$ converges to 3 while $\{b_n\}$ converges to -1. Then the sequence $\{a_n - b_n\}$ converges to (1)
- A. 1
B. 4
C. 3
D. $1/3$
-
34. Which of the following argument is valid? (1)
- A. $p \vee q, p \Rightarrow q \vdash \neg q$
B. $p \Rightarrow q, q, r \Rightarrow p \vdash r$
C. $p, p \Rightarrow q, r \vee \neg q \vdash p \wedge r$
D. $p \Rightarrow q, \neg r \wedge q \vdash p \vee r$
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35. Which of the following assertions is true applying the Principle of Mathematical Induction? (1)
- A. $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for $n \geq 0$
B. $n! \geq 2^n$ for $n \geq 0$
C. $2 + 3 + 4 + \dots + n = \frac{n(n+1)}{2} + 1$ for $n \geq 2$
D. $3 + 5 + 7 + \dots + (2n + 1) = n^2$ for $n \geq 1$
-
36. Which of the following is equal to $\int 3\cos 3x dx$? (1)
- A. $\frac{1}{3} \sin 3x + c$
B. $\sin 3x + c$
C. $\sin x + c$
D. $-\sin x + c$
-
37. What is the area of the region bounded by the graph of $f(x) = x^2 - 4$ and the x-axis in square units? (1)
- A. 16
B. 32
C. $16/3$
D. $32/3$
-
38. Which of the following set of equations define the xz-plane? (1)
- A. $\{(x,y,z) | x,y,z \in \mathbb{R} \text{ and } x=9\}$
B. $\{(x,y,z) | x,y,z \in \mathbb{R} \text{ and } y=0\}$
C. $\{(x,y,z) | x,y,z \in \mathbb{R} \text{ and } x=0\}$
D. $\{(x,y,z) | x,y,z \in \mathbb{R} \text{ and } z=0\}$

39. Which of the following functions has a graph that completes one full cycle on $[-\pi, \frac{11\pi}{4}]$? (1)
- A. $f(x) = 3 \sin(3x + \frac{\pi}{6})$ B. $f(x) = 3 \sin(\frac{4}{3}x + \frac{\pi}{2})$
 C. $f(x) = 2 \sin(\frac{2}{3}x - \frac{\pi}{6})$ D. $f(x) = 2 \sin(2x - \frac{\pi}{6})$
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40. Let A be a square matrix of order n and let B be a matrix obtained by multiplying one row of A by 2 and C be a matrix found by adding one row of A to another row of A. If $\det(A) = 10$, what are $\det(B)$ and $\det(C)$? (1)
- A. 20 and 10 B. 20 and 0
 C. 10 and 20 D. 10 and 10
-
41. The solution set of the system $\begin{cases} 2x - y + 3z = 1 \\ x + y + 3z = 5 \\ x - y + z = -1 \end{cases}$ is (1)
- A. $\{(-4 - 2k, k, 2 - k) : k \in \mathbb{R}\}$ B. $\{(k + 1, 2k, k - 1) : k \in \mathbb{R}\}$
 C. $\{(2 - 2k, 3 - k, k) : k \in \mathbb{R}\}$ D. $\{(k, k + 1, k - 1) : k \in \mathbb{R}\}$
-
42. Let z_1, z_2 , and z_3 be complex numbers. Then which of the following is NOT necessarily true? (1)
- A. $z_1 - (z_2 + z_3) = (z_1 - z_2) - z_3$ B. $z_1 - (z_2 - z_3) = (z_1 - z_2) - z_3$
 C. $(z_1 z_2) z_3 = (z_2 z_1) z_3$ D. $(z_1 z_2) z_3 = z_1 (z_2 z_3)$
-
43. If a translation T takes a point $(-1, 1)$ to a point $(3, 2)$, then the image of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$ under T is (1)
- A. $(x - 3)^2 + (y - 3)^2 = 4$ B. $(x - 2)^2 + (y + 3)^2 = 4$
 C. $(x - 5)^2 + (y + 1)^2 = 4$ D. $(x + 3)^2 + (y + 3)^2 = 4$
-
44. Let $u = (-1, 6)$ and $v = (2, 4)$ be position vectors. Then the vector $4u - 3v$ is (1)
- A. $(-10, 12)$ B. $(-10, 11)$
 C. $(-4, 24)$ D. $(-15, 18)$
-
45. Water is running into a conical tank, which is pointed down at a constant rate of $6\pi \text{ cm}^3/\text{sec}$. The cone has a height of 10 cm and a base radius of 5 cm, then how fast is the water level rising when the water reaches 2 cm deep? (1)
- A. $\frac{4}{\pi} \text{ cm/sec}$ B. 6 cm/sec
 C. 4 cm/sec D. $\frac{6}{\pi} \text{ cm/sec}$
-
46. Let F be the anti-derivative of f. Then which of the following is NOT true about F and f? (1)
- A. $\frac{d}{dx} \int F(x) dx = f(x)$ B. $F'(x) = f(x)$
 C. $\int F(x) dx = f(x) + c$ D. $\int \frac{d}{dx} f(x) dx = f(x) + c$
-
47. Let $f(x) = \frac{1}{\sqrt{x}}$ and $g(x) = 2 - \frac{1}{6}x^3$ be two functions. For what values of x do the graphs of f and g have tangent lines of the same slope? (1)
- A. $\frac{\sqrt{3}}{2}$ B. 1
 C. 4 D. 1/4
-
48. What is the value of a number in the interval $(0, 1)$ that satisfies the Mean Value Theorem for the function $f(x) = 3x^2 - 5x + 1$? (1)
- A. 1/2 B. 3/5
 C. 5/6 D. 1

49. Let $f(x) = 3x + 2x - \sin x + e^x$. What is the value of $f'(0)$? (1)

- A. 2
B. 1
C. -1
D. 4

50. Let f and g be differentiable functions, and $h(x) = g(f(3x + 1))$. If $f'(1) = f(1) = 2$ and $g'(2) = 1$, then what is the value of $h'(0)$? (1)

- A. 2
B. 1
C. 6
D. 4

51. Which of the following describes the mechanical meaning of derivative? (1)

- A. Gradient of graphs of functions
B. Area under the curve
C. Instantaneous rate of change
D. Slope of tangent line

52. Which of the following is true about the zero(s) of a function $f(x)$? (1)

- A. If $f(a)=0$, then $x-a$ is a factor for $f(x)$.
B. If $f(a)=0$, then $(x-a)^2$ is a factor of $f(x)$.
C. If $f(a)=0$, then a is not necessarily in the domain.
D. If $f(a)=0$, then $y=a$ is they-intercept.

— END OF EXAMINATION —

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